



Welcome! This alphabetized glossary contains many of the terms you'll find within this course. This comprehensive glossary also includes additional industry-recognized terms not used in course videos. These terms are important for you to recognize when working in the industry, participating in user groups, and participating in other certificate programs.

Terms	Definitions
Access control	Access control prevents unauthorized viewing, modification, or copying of data. IT staff use access control to restrict what users can do, which resources they have access to, and what functions they are allowed to perform. (See RBAC and ROLP.)
Access tokens	Bits of code that signify a user has successfully authenticated their identity. Hackers get access tokens by sniffing network traffic between trusted entities. Once they find an access token, they can hijack the session and use the token to impersonate the trusted entities. After that, the hacker can intercept and modify any information sent or access private accounts as if they were the account holder. (See Replay attack.)
ActiveX	Old online browser technology for downloading and embedding content on websites.
Add-ons	Small bits of code or software applied to browsers to enhance functionality and user experience. Add-ons have evolved over the years. Toolbars, plug-ins, and extensions are all examples of browser add-ons.
Adware	Software coded into online ads that records your personal data, website visits, and keystrokes to send you personalized ads. Adware can be legitimate or malicious. Adware is often bundled with apps, it activates on install. Adware can collect data, track online activity, or link out to malicious websites with viruses. It can also cause sluggish system performance, crashes, changes to your browser's home page, the appearance of new extensions, toolbars, or software, and web pages not displaying properly. Popup ads and the sudden appearance of unfamiliar apps are the most common way to recognize adware.
Anti-replay protection	Network standards that stop hackers from re-using data (See IPsec.)
Asymmetric encryption	Uses a public key and a private key. It takes longer than symmetric encryption because it is more complex. It is used for smaller amounts of data. It is safe to widely share the public key for encryption or decryption because only the secret key can undo the public key's action. It's used in Authentication, digital certificates, digital signatures, and key exchange – where a symmetric encryption key is shared only to specific recipients. Also called "Public Key Cryptography". (See Symmetric encryption.)
Authentication	The act of confirming the identity of a user. Authentication involves two steps: entering the correct login information and confirming that it is really you.

Authentication factors	Ways to confirm a user's identity. These include something you know, something you have, and something you are. Examples include the answer to a security question, a 2FA security card, and a fingerprint scan.
Authentication header protocol (AH protocol)	Protocol that authenticates the sender and both IP addresses (See IPsec.)
Authentication methods	Ways to log in to a system. These include SSO, 2FA, and MFA.
Authorization	Permission to access a location or do an action. Access control must be set up before authorization can be granted.
Autorun	Computer feature that allows inserted drives and disks to run or play automatically. Disabling autorun is recommended since it would allow an infected drive to install malware automatically the moment it is inserted.
Baiting	In social engineering, the use of a lure just like the bait or lure you'd use to catch a fish. Lures can be physical or logical. (See Social engineering.)
Basic Input Output System (BIOS)	Firmware that boots up Windows and Linux PCs, runs hardware checks, and starts the OS. (See Firmware.)
Basic Input Output System password (BIOS password)	A security feature that requires a correct password in order to start a system's OS. Also called 'firmware password.' (See Firmware and BIOS.)
Biometrics	Body scan that can confirm whether a person physically accessed a device, network, or area. (See Non-repudiation.)
Bloatware	Unwanted trial software, usually pre-loaded on new devices. Aging bloatware is soon outdated and may have vulnerabilities.
Browser history	A list of every page you've visited and for how long. Browser histories help you find sites you visited, but they can also be used against you by family, hackers, or law enforcement. Use browser settings to clear history.
Browsing history	List of recently visited web sites. Anyone with access to your device can see what sites you visited. Attackers use browsing history to learn where they might impersonate their victims, and companies use it to see which sites you go to on your work computer. (See Digital accounting.)
Brute force attacks	A type of password cracking attack where hackers submit as many passwords as possible, hoping one will work.
Buffer overflow	Common type of DoS attack that sends a website more Internet traffic than it can handle.
Cache	Storage area that holds downloaded web pages you've visited. Cached pages load immediately. If a page has changed since the last cache, it is downloaded, displayed, and cached again. Over time, very large caches cause sluggish performance. Use browser settings to clear the cache, set disk space limits, and set expiration dates.

Certificate authority (CA)	A trusted organization that issues digital certificates. The major CAs are Apple, Microsoft, Mozilla, and Google. Which CA your device uses depends on the operating system it's running. Also called 'Root certificate store'. (See Digital certificate, SSL certificates, and HTTPS websites.)
CIA Triad	Confidentiality, integrity, and availability. Confidentiality means that data is protected from unauthorized access. Integrity means that data is protected from unauthorized changes. And availability means that you have access to your data whenever you need it.
Cipher	An algorithm that is used to scramble plain text, so it is secure.
Ciphertext	Plain text that is encrypted, which makes it unreadable.
Company Confidential Information	Any information that is used to run a company, like intellectual property, product designs, procedures, plans, employee records, and financial data. (See Confidential information.)
Confidential information	Information that must be kept secret. Employees are trained to recognize and deal with confidential information so that it remains secure. Companies rank information and files by how sensitive each one is. Each company ranks their information differently, but there are four main types of confidential information that should be universally protected: PII, Company Confidential information, Customer Confidential Information, and PHI.
Cookies	Text files with small pieces of data. When you visit a site, the server creates a cookie and saves it on your browser to track you. Cookies let websites remember your logins, shopping carts, and more. This creates an easier, more personalized online experience. Cookies can also be used to ban you from a website if you've violated any of its conditions for use. Most cookies are safe, but some are designed to track without consent. Even legitimate cookies can be harmful if you get hacked. (See Digital accounting.)
Cookies, authentication	Authentication cookies save logins, usernames, and passwords, so you don't have to remember them.
Cookies, session	Session cookies are used only for one session. They are stored in RAM and are automatically deleted when the browsing session ends.
Cookies, third-party	Third-party cookies are from sites you are not on. These track you across the web. Ads can generate cookies, even if you never click on them.
Cookies, zombie	Zombie cookies are third-party cookies that don't follow normal cookie protocols. They can store their code directly on your device to recreate themselves even after deletion. They are extremely difficult to remove. Websites may also use zombie cookies to ban specific users.
Cryptographic hash	A short string of numbers and letters created by running a password or file through an algorithm. A single password and a full library will have different cryptographic hashes, but each hash will have the same number of characters. If any data is altered or removed from a password or file, its cryptographic hash will be different. When you create a password, it's converted into a cryptographic hash. On your next log in, it's converted again. If the stored hash and the new hash match, the system lets you in. Cryptographic hashes save space, authenticate data, and keep information secure.

Customer Confidential Information	Information customers or partners provide to companies, which includes PII and also things like purchase histories, and credit card information. (See Confidential information.)
Data	Raw values and facts are usually collected by automated systems. For example, page visits, link clicks, and monthly sales. (See Information, Insights, and Data analytics.)
Data analytics	The processing of raw data, like values or facts to create meaningful information. (See Information, Insights, and Data.)
Data at rest	Data that resides on a storage device. The files aren't open or being transmitted anywhere. Data at rest can be encrypted at the file level, storage device level, and cloud level. Encryption at the file level lets you encrypt all your files on a storage device or just a select few. Data at rest is less vulnerable than data in motion, but it's not immune from attacks.
Data availability	Data availability means that authorized users have immediate and reliable access to their data. This includes granting access to authorized users with passwords and security questions. Data availability can be compromised by hardware failures, unscheduled software downtime, network bandwidth issues, cyberattacks and sabotage. To protect against these threats, systems that require high availability (99.999% uptime) have network monitoring, redundant systems, and backup servers ready to take over. If the primary system is compromised, business continuity and customer access can be maintained.
Data breaches	Data breaches are when a data leak is caused intentionally by a cybercriminal. These occur when social engineering or phishing attacks trick employees into leaking sensitive credentials or information.
Data capture	The collection of data from multiple sources and the secure storage of it in relational databases, or more commonly, semi-structured data warehouses. Data may be captured by server logs showing where customers browse, IoT sensors in home appliances and business technology, or customer and employee surveys or rating systems. (See Data correlation and Meaningful reporting.)
Data confidentiality	The control of data access and the use of security tools like encryption and multifactor authentication (MFA) to keep data secure. Data confidentiality prevents identity theft, compromised accounts and systems, legal concerns, damage to reputation, and other severe consequences. To determine if data should be confidential, ask: Who is authorized? Do confidentiality regulations apply? Are there conditions for when data can be accessed? What would the impact of disclosure be? Is the data valuable?
Data correlation	When raw data points are analyzed to find connections or links. For example, Netflix uses tools that compare searches, views, and ratings so they can predict which movies and shows will be successful on their platform. AI and machine learning algorithms automate parts of the analysis. (See Data capture and Meaningful reporting.)
Data dumps	Data dumps are when cybercriminals dump stolen data onto the dark web for monetary gain. A data dump might include PII, PHI, bank account numbers, PINs, social security numbers, and more. Other cybercriminals buy and use data dumps for things like identity theft and password attacks.

Data in motion	Data that is actively moving between two devices – meaning two computers, a mobile device and a mail server, or your computer and your bank's online website. Data in motion is especially at risk for interception attacks like man-in-the-middle. Also called data in transit.
Data in transit	(See data in motion.)
Data integrity	The collection of actions and fail safes that protect data. Data integrity guarantees that data is accurate, complete, and consistent. It covers data in storage, during processing, and in transit. The two main types of data integrity are physical and logical. Without data integrity, loss, corruption, or compromise can cause significant damage and financial loss for both businesses and customers.
Data integrity (logical)	Logical data integrity are the checks and protocols that protect data from human error and hackers. These confirm that data is correct and accurate as it's used in different ways within an organization.
Data integrity (physical)	Physical data integrity is the collection of actions and fail safes that protect the physical systems that store and process the data.
Data leaks	The accidental exposure of confidential or sensitive data through a security vulnerability.
Data-driven business decisions	Business decisions based on data capture, data correlation, and meaningful reporting.
Distributed denial of service attack (DDoS attack)	DoS attack made with a large collection of compromised, malware-infected computers known as a botnet. It's harder to identify a DDoS attack's origin, which makes it harder to shut down. And DDoS attacks are far more devastating than DoS attacks since hundreds or thousands of computers are used instead of just one.
Decryption key	(See Encryption key.)
Default usernames and passwords	The default login information that comes with new hardware or software. These are essential to tech support, software installation, and device configuration. They have admin-level privileges and hide who's using them, as they aren't associated with any registered users. Hackers use them to break into apps, devices, OSes, databases, and BIOS since they are easily found online in help guides or user manuals.
Denial of Service (DoS)	Cyberattack, where a website or server is targeted with so much traffic that it overwhelms that system, degrading performance until the server is unable to respond.
Denial of service attack (DoS attack)	Cyberattack that floods a network with so much traffic that it crashes. DoS attack victims are typically high-profile, like government sites, banks, or social media sites. Sometimes, DoS attacks are used to distract from other attacks happening at the same time.
Device lock	The literal locking of a device—either physically, behind a locked door, or with a steel cable and padlock; or digitally with passwords or PINs.
Dictionary attacks	A type of password cracking attack where hackers use words pulled from dictionaries or newspapers to crack passwords.

Digital accounting	Investigating online and network activity to discover the reasons behind a certain outcome. Digital accounting is used in troubleshooting, security analysis, forensics, and hacking.
Digital certificate	A security approval that includes a public encryption key that encrypts data. If the data recipient trusts the CA that issued the digital certificate, they use a private key to decrypt the data. Digital certificates are also used in smart card authentication. (See CA and PKI.)
Digital Millennium Copyright Act (DMCA)	Law that makes it illegal to bypass copy protections or to develop technology that helps bypass copy protections. (See DRM and Digital products.)
Digital products	Non-tangible assets a company owns, like software, online music, online courses, e-books, audiobooks, and web elements like WordPress templates or Shopify themes. (See DRM and DMCA.)
Digital receipt	Encryption and authentication method used to prove that a message was sent from one party to another. (See Non-repudiation.)
Digital Rights Management (DRM)	Code added directly to files that helps prevent digital assets from being copied or pirated, but there are tools that can remove DRM code. (See Digital products and DMCA.)
Digital signatures	A regular signature used in conjunction with a hardware or software token. This authenticates the signer. The sender sends an encrypted signature and a public decryption key to a recipient. If the recipient can decrypt the signature with the public key, that proves the sender signed it because they must have performed the encryption with the private key. (See PKI and Non-repudiation.)
Drive encryption	The scrambling of a drive's data so it's unreadable. (See Firmware.)
Dumpster diving	Dumpster diving is the act of physically searching through a literal dumpster to find something valuable. A company's trash might contain lists of customer names, phone numbers, contact information, business plans, product designs, or an access code written on a post-it note. Tech companies require document shredding and device destruction as a normal part of business because these can be stolen from the trash to harvest data that can be used for identity theft and data breaches. Or the data could be sold to hackers or a company's competitors.
Eavesdropping	When hackers use a packet sniffer to read unencrypted network traffic so they can intercept, alter, or delete data transmitted between devices. Eavesdropping (or packet sniffing) attacks occur on wireless, wired, and phone connections. Staying off public wifi or using encryption with a VPN or cellular connection helps prevent eavesdropping attacks. Eavesdropping is also called "packet sniffing". (See Packet sniffer.)
Email management	Classifying email messages and deciding whether they should be saved or deleted. It helps prioritize emails, saves time, and increases productivity. It includes using folders and subfolders, using rules or filters, unsubscribing from email lists, and configuring settings to block spam.
Encapsulating security payload	Protocol that encrypts data and authenticates data and senders. (See IPsec.)

protocol (ESP protocol)	
Encryption	Encryption is the act of taking readable plain text and scrambling it into unreadable ciphertext with an algorithm, so it can only be read by a recipient that has the decryption key. Encryption is used at the network layer for data traveling across networks. It can also be done locally to hard drives, phones, and even thumb drives so that lost device data remains unreadable. In some industries, data encryption is a mandatory requirement. This includes student records, medical records, and consumer data. Many OSes have encryption built in. For those that don't, third-party encryption software is available.
Encryption and hashing algorithms	Algorithms that scramble data. (See IPsec.)
Encryption key	A series of random, unique numbers combined with very powerful algorithms that are used to encrypt (or scramble) data before it is sent. The person on the receiving end has a decryption key that's used to decrypt (or unscramble) the data, so it's in a readable or usable format.
Exploits	Pieces of code that use vulnerabilities in hardware or software to get into a system. Malware-infected websites use exploits to automatically download malware to a system. This is called a drive-by download.
Extension	Small piece of source code that adds a function or feature to a browser. Ad-blockers and in-browser PDF readers are extensions. Since extensions are given special authorizations within the browser, they are attractive targets for attackers. (See Add-ons.)
File integrity monitoring (FIM)	Software that audits sensitive files and folders to ensure all activity is authorized.
Firewall	Software or hardware that monitors connections and blocks harmful traffic based on preset rules. For example, schools and businesses use firewalls to block social media sites, age-inappropriate content, and certain types of downloads.
Firmware	Firmware is software that tells hardware how to behave. Security firmware protects devices and data from malware and tampering. Outdated firmware leaves devices vulnerable.
Firmware password	A security feature that requires a correct password in order to start a system's OS. Also called 'firmware password.' (See Firmware and BIOS.)
General Data Protection Regulation (GDPR)	Set of regulations that mandate digital privacy for all countries in the European Union.
Hardening	The process of securing a device to minimize vulnerabilities. This includes disabling unneeded device features, regularly updating a device's firmware, OS, and software, and using firewalls, VPNs, and antimalware. (See Patches.)
Hashing	Hashing is when an algorithm transforms an input string (like your password) into a smaller, fixed-length output string (or hash) that's saved to a file. A hash is like a digital fingerprint. Passwords are hashed with a scrambling algorithm. If a password hash is determined, attackers can use it to determine other passwords that were

	scrambled in the same way—that can be over 90% of unknown passwords in some cases. (See Password hash.)
Health Insurance Portability and Accountability Act (HIPAA)	Set of regulations that mandate the use and disclosure of protected health information in America.
Hypertext Transfer Protocol Script protocol (HTTP protocol)	A protocol that governs the flow of online traffic.
HTTP websites	Websites relying on the HTTP protocol are not secure, especially when visited over public wifi. Everything that you send and receive is in plain text. It makes you an easy target for cybercriminals. (See HTTPS websites.)
Hypertext Transfer Protocol Script Secure protocol (HTTPS protocol)	A protocol that governs the flow of online traffic and provides encryption for security.
HTTPS websites	Secure HTTP, or HTTPS, protocol provides an encrypted connection between you and the sites that use it. Most commercial websites, social networking sites, or sites that offer a customer login use HTTPS. The easiest way to tell is to see if a website is HTTPS is if the URL starts with HTTPS. Or you can look for a lock icon in your browser's URL window to confirm that site's digital certificate and other security information. HTTPS websites only provides a secure connection between you and the web server. A site can use HTTPS and still try to scam users or be compromised in some way. (See HTTP websites and SSL certificates.)
ICMP flood	Common type of DoS attack where diagnostic pings are sent to every computer on a network. Each computer pings every other computer, and so on, until the network crashes.
Identity fraud	The act of using the personal, private, or financial information stolen from victims to commit fraud. (See Identity theft.)
Identity theft	The act of stealing personal, private, or financial information from a person with the intent of using it to assume the victim's identity and to commit fraud. (See Identity fraud.)
Impersonation	When a hacker sets up a public wifi network that seems legitimate. Once a user connects, login credentials, session information, and PII can be intercepted. Or when a hacker sets up a fake website that looks and feels exactly like a real website, such as a well-known bank or other high-profile site. They may send fake email or text links (known as phishing) to trick you into visiting the fake site so they can steal your credentials to the real site and install malware on your device. A third type of impersonation is when a hacker pretends to be someone else so they can steal data or take over systems. This is also called social engineering. Impersonation attacks can be used individually or in combination with each other.
Incognito mode	(See Private browsing.)

Information	A summary of raw data. For example, positive or negative results that happen after some specific change. (See Data, Insights, and Data analytics.)
Information asset	Information or data that has value. Like patient records, customer information, and intellectual property. Information assets can exist physically, on paper, on disks or other media, or they can exist electronically in databases and files.
Insights	Conclusions based on the results of information analysis. Meaningful business decisions are based on insights. For example, if a positive trend occurs after store hours are changed, the right business decision would be to maintain those new hours. (See Information, Data, and Data analytics.)
Intellectual property (IP)	Creations of the mind that are generally are not tangible. Often protected by copyright, trademark, and patent law. Examples of IP include industrial designs, trade secrets, research discoveries, and even some employee knowledge. Companies use a legally binding document called a Non-Disclosure Agreement (NDA) to prevent the sharing of IP and other sensitive information.
Internet key exchange (IKE)	Secure exchange of cryptographic keys. (See IPsec.)
Internet Protocol Security (IPsec)	A suite of network standards and protocols that use cryptography to protect data traveling over the Internet. The core protocols for the IPsec suite are the AH and ESP protocols. To support data security, the IPsec suite uses security associations (SA), Internet key exchange (IKE), encryption and hashing algorithms, and anti-replay protection. IPsec has two modes: tunnel mode and transport mode.
IPsec transport mode	In IPsec transport mode, the IP header of the original data packet is left unencrypted. Only the content of the data packet is encrypted. Transport mode is common in host-to-site VPN. (See IPsec.)
IPsec tunnel mode	In IPsec tunnel mode, the entire data packet is wrapped in a new packet, encrypted, and given a new header. Tunnel mode is common in site-to-site VPN. (See IPsec.)
Jailbreaking	When users remove device restrictions on Apple iPhones so they can make changes and install non-approved apps. This adds functionality but also adds vulnerability. It is harder to jailbreak newer model phones, and there is less reason for it with updated features and app selections. (See Rooting.)
Java	Programming language used to enhance website interactivity.
Logical lure	An online ad or social media platform that looks too good to be true so users will click the ad to find out more. It's a set up so an attacker can infect their computer with malware to gain access to login credentials, financial information, and other valuable data. (See Social engineering and Baiting.)
LogsFiles	that capture access, changes, error messages, and other basic information. Most software and systems generate audit logs. Audit logs capture log file events which can show who did what and how the system behaved. (See Digital accounting.)
Macro viruses	Bits of code that affect Microsoft Office files via the macros they use to automate tasks.

Malicious websites	Websites that mimic real websites to trick users into entering login or payment data or to install malware that gathers personal information or takes over a device. Malicious website pages often have typos or designs that don't match the legitimate site they are mimicking. Their URLs might use a zero instead of an "O" or a capital "i" instead of a lower case "L".
Malware	A general term for software designed to compromise computer systems. Malware can cause system slowdowns, odd requests, browser misdirection, and popup ads. It can also steal data, record everything you do with or near your device, spam your contacts with infected links, and connect your computer to a network of hijacked computers that are remotely controlled (known as a botnet). Malware can come from attachments, sketchy websites, file downloads, infected USB drives, or links in emails, ads, social media, torrents, and even text messages. Malware types include viruses, worms, trojans, exploits, spyware, adware, and ransomware.
Man-in-the-middle attack	A man-in-the-middle attack is a form of eavesdropping. It has a victim, a receipt point, and an attacker. The victim and receipt point are unaware the attacker is listening in. Man-in-the-middle attacks can be physical or logical. Other man-in-the-middle attacks include spoofing, hijacking, and theft of browser cookies. (See Eavesdropping.)
Man-in-the-middle attack (logical)	In a logical man-in-the-middle attack, the attacker sends emails or texts with fake links that direct victims to sites that steal their data and install malware. If a fake email warned about a bank account problem, the victim might click the link and try to login. This gives the attacker control of their bank account and installs malware on the victim's computer.
Man-in-the-middle attack (physical)	In a physical man-in-the-middle attack, the attacker is physically near the victim, like the same public wifi network or a network they set up themselves as a trap. The attacker sniffs the unencrypted network traffic to gain access to everything the victim is doing online so they can steal information.
Meaningful reporting	The presentation of analyzed information in ways that help people further analyze and interpret. Reporting tools use captured and correlated data to provide charts, keyword searches, and graphs that help companies achieve business insights. (See Data correlation and Data capture.)
Multi-factor authentication (MFA)	Authentication method that requires multiple authentication factors before authentication is granted. Usually, MFA requires that you fill in a code sent to your email or phone to prove that a login attempt came from you.
Near field communication (NFC)	Wireless technology that transfers data across devices with a tap or a bump. It's usually used for payments or sharing contacts. It has a much shorter range than Bluetooth and has zero security protections aside from its limited range.
Non-repudiation	When you can't deny being in a specific location. It guarantees that a message sent between two parties is genuine. Non-repudiation examples include video, biometrics, digital signature, and digital receipt. (See Non-repudiation.)
Open network	Free, unencrypted networks that do not require a password to log on. Airports, coffee shops, hotels and even fast-food restaurants offer their guests access to open networks. Open network providers may track what you do and sell your data to advertisers. Open networks also invite eavesdroppers to view everything sent and

	received on that network. On unprotected devices, they can install malware or hijack user sessions. Also called 'unsecure network' and 'public wifi'.
Original equipment manufacturer (OEM)	The original maker of a hardware device. OEMs provide drivers and firmware updates on their websites.
Packet sniffer	A packet sniffer is a tool that intercepts everything transmitted on a network. Anything your device sends on an unencrypted network can be viewed with a packet sniffer. If a network is encrypted, packet sniffers will only be able to see things like the origin and destination of a packet but not the data inside it. (See Eavesdropping.)
Packet sniffing	(See Eavesdropping.)
Password cracking	Getting a correct password in an unauthorized way.
Password hash	The string of letters and numbers that result after a password is processed by a scrambling algorithm. (See Hashing.)
Patches	Patches are updates to apps and OSes that fix security weaknesses. Companies regularly release patches alongside system improvement updates to make sure that their customers are safe from new threats. But patches are a response to KNOWN threats—meaning the threat has already happened to someone. (See Hardening.)
Personally Identifiable Information (PII)	Any information that can be used to identify someone, like government ID numbers, birthdates, addresses, and phone numbers. (See Confidential information.)
Phishing	Email or text-based social engineering attacks that hackers use to steal usernames and passwords, bank account information, Social Security numbers, and more. Phishing attacks make it look like the email or message came from a friend or family member, or someone official like a bank, the government, or a large company. They use fear, greed, or a sense of urgency, so users are more likely to make a rush decision and click a malware-infected link or attachment. For example, a data breach alert from your bank that includes a password reset link. The email message and website were fake, and your password was never reset. Phishing emails usually have typos and grammatical errors. (See Social engineering.)
Physical lure	A physical object, such as a malware-infected USB flash drive that's been deliberately left somewhere in the hope that someone will take the drive and plug it into their computer. It's a set up so an attacker can gain access to login credentials, financial information, and other valuable data. (See Social engineering and Baiting.)
Plain text	Data that has not been encrypted.
Plug-ins	Site-specific browser add-ons that you click on to install. They are not supported in today's modern browsers, in favor of extensions. (See Add-ons.)
Polymorphic viruses	Bits of code that change their characteristics to get around cybersecurity defenses. 97% of all malware uses polymorphic viruses.
Pretexting	When an attacker poses or impersonates someone with authority – such as a police officer, someone from your bank, a tax official, or one of your co-workers. They can use a sense of fear, intimidation, friendliness, sympathy, or urgency to try to trick or pressure you into confirming your identity under the pretense that they need

	something from you or need you to perform some critical or urgent task for them. (See Social engineering.)
Private browsing	Browser setting that doesn't save browsing history, cookies, site data, or form data. Also called 'incognito mode'. Private browsing users appear as a new or unknown user on sites, and other people who use the device won't see private browsing history, and cookies and site data are remembered while browsing but deleted when the browser is closed. But private browsing activity isn't hidden from employers, schools, or ISPs. Also, any bookmarks or downloads created during private browsing will be kept.
Program viruses	Bits of code that insert themselves into another program.
Protected Health Information (PHI)	Any information added to a person's medical record during diagnosis or treatment that can be used to identify them, like PII, medical history, prescription lists, photos, and more. (See Confidential information.)
Public key infrastructure (PKI)	When a user is validated with a digital certificate by a Certificate Authority. (See Smart card authentication, Digital certificate, and Digital signatures.)
Public wifi	(See Open network.)
Rainbow attacks	A type of password cracking attack where hackers use words from an original password hash to generate all other possible passwords.
Ransomware	Software that locks a system, encrypts its files, and displays a ransom demand. To get the encryption key, you must pay the ransom. Or you can regain access by doing a full system restore from a backup.
Relational database management system (RDBMS)	A database that records user access and data changes. RDBMS is safer than a spreadsheet program.
Replay attack	A type of man-in-the-middle attack which intercepts and retransmits data. Replay attacks involve "trusted entities" and require an "access token". Replay attacks are also known as repeat or playback attacks.
Role-based access control (RBAC)	Access control method that follows a company's org chart. Different customer and employee roles are set up as groups on a network, and then those groups are granted certain permissions. When a new user joins the network, they are assigned to the group that fits their role. They will have the lowest level of permissions they need to do their job. (See Access control and ROLP.)
Root certificate store	(See CA and SSL certificates.)
Rooting	When users remove device restrictions on Android phones so they can make changes and install non-approved apps. This adds functionality but also adds vulnerability. It is harder to root newer model phones, and there is less reason for it with updated features and app selections. (See Jailbreaking.)
Rule of least privilege (ROLP)	Access control method where access is only granted to resources that a user needs to fulfill their role. (See RBAC and Access control.)

Search engine algorithms	Algorithms used by search engines that detect harmful sites. Browsers use those results to warn users that a site may contain malware, malicious code, phishing scams, or be hacked in some other way. If your browser tells you a site is unsafe, avoid that site.
Secure boot	A feature of UEFI. It confirms an OS manufacturer's digital signature, which prevents malware from taking control during boot-up. (See Firmware and UEFI.)
Secure sockets layer certificates (SSL certificates)	Certificate that authenticates a website's identity and enables an encrypted connection between a web server and a browser. SSL certificates come from CAs. When you browse a website using HTTPS, you're trusting the CA to validate the information submitted by the business who has requested an SSL certificate. Businesses need SSL security certificates for their websites so they can keep user data secure, verify their ownership of the site, prevent attackers from creating fake versions of their site, and to convey trust to users. They also need them if they want to have an HTTPS web address. When you see a lock icon in the address bar of a URL you visited, then you know that website is using HTTPS. Also called 'security certificate' and 'trust seal'. (See CA and HTTPS websites.)
Security associations (SA)	Defines which types of hashing and encryption are used (See IPsec.)
Security certificate	(See SSL certificate.)
Security zones	A feature of older browsers. They block or allow websites and confine them to different zones such as the local network, Internet, or intranet. For each zone you could apply different security levels. For example, medium, high, or custom. Security zones can also be configured to allow ActiveX and Java for added website interactivity or functionality, but this introduces risk. Modern browsers do not use Zones. Older web apps may require you to enable ActiveX in order to run properly.
Session hijacking	When an attacker intercepts and hijacks your connection. For example, they can take over a connection while you're logged in to your bank and transfer funds to an outside account.
Shoulder surfing	When an attacker sits strategically in public areas to try to see usernames and passwords, bank account pins, and other essential information. They literally watch what you enter on your screen and look at your finger's keystrokes, too. Shoulder surfing is common in busy public places.
Smart card authentication	User authentication using a card that has a security chip. The smart card has a public/private key pair. It presents a digital certificate (including the public key) to the server it's trying to access. If the server trusts the CA that issued the digital certificate, it will use the public key to send an encrypted request. Only the smart card's private key can decrypt the request, which means only the smart card owner can send the correct response. (See PKI and digital certificate.)
Snooping	Cyberattack that intercepts data between devices. Snooping can reveal logins, credit card numbers, intellectual property, and more. Snooping attack types include: eavesdropping, man-in-the-middle, and replay. Some hackers can even use a computer monitor's electromagnetic fields to reconstruct what it displays. Snooping is common on open, unsecured networks and can be difficult to trace.

Social engineering	The use of psychological manipulation to trick people into giving away sensitive information or getting them to make security-related mistakes. Attackers try to gain a victim's trust and then try to manipulate that person into doing what the attackers want them to do. Often times, the sense of fear or urgency is used to further the process along. Social engineering is a trap. It relies on human error. (See Pretexting, Baiting, and Phishing.)
Software or license theft	The unauthorized copy or use of copyright-protected software. This includes pirating software and counterfeiting activation codes.
Spam	Unwanted, unsolicited digital communication sent out in bulk to multiple recipients at once. Much of the time, it is sent from unknown senders. It comes in the forms of email, text messages, instant messages, robocalls and social media. It is junk, typically used for advertising products, goods, or services, but it can also be used to distribute malware. Your company and web-based email services filter out most spam at the email server level and block domains known for sending spam. But some still gets through. Some spam is harmless, but it can be dangerous when scammers use it to commit phishing attacks or fraud against you.
Spyware	Software collects personal data, login credentials, credit card information, online activity, and can record using a device's camera or microphone. Spyware can be legitimate or malicious.
Single sign-on (SSO)	Authentication method that lets you log in to multiple applications and platforms with one login.
Stealth viruses	Bits of code that copy themselves to different locations to avoid antivirus scans.
Strong password	Passwords that are long, difficult to guess, and have a mix of numbers, letters, symbols, and capitalizations. For example: the phrase, "Johnny Appleseed loves apples" would be a very strong password, especially if it included a mix of numbers and symbols.
Symmetric encryption	When a single key is used between parties to encrypt and decrypt data. With only one key, it uses less memory, which is great for quickly and securely processing larger amounts of data. That's why it is often used to protect the main data exchange in a session. But it's harder to keep a single key secret, especially if it needs to be broadly distributed. If this key is intercepted by a hacker, then they can decrypt messages, hack accounts, and steal or tamper with data. 3DES and CAST are examples of symmetric encryption technologies. Also called "single-key" or "private key" encryption. (See Asymmetric encryption.)
SYN flood	Common type of DoS attack that sends a rapid series of incomplete connection requests which flood a website until the server crashes.
Toolbars	Toolbars were designed to add functionality to the browser, like spell check, auto fill, yellow pages, and dictionaries. But most toolbars were considered nuisance-ware. Some would show sponsored results instead of what you originally searched for, or they'd impede system performance. Some even installed malware, tracked browsing habits, and collected user data. Toolbars used to be forcibly installed on user devices during software installation processes. Fortunately, toolbars are a thing of the past. (See Add-ons.)

Tracking	Following a person to see where they go and what they do. Websites can track your OS, browser version, installed extensions, screen resolution, installed fonts, time zone, language, and how long you spent on a site and what you did there. (See Digital accounting.)
Trojans	Bits of code that trick you into installing legitimate-seeming software that includes harmful malware.
Trust seal	(See SSL certificate.)
Trusted entities	Users or websites that get an access token (or security key) after verifying that they are who they say they are. For example, connecting to your bank or your work network on a network-registered device. (See Replay attack.)
Trusted Platform Module (TPM)	A chip that stores and manages encryption keys. TPM chips won't start a device or unencrypt data if tampering is detected. (See Firmware.)
Two-factor authentication (2FA)	Authentication method that requires an authentication device before authentication is granted.
Unified Extensible Firmware Interface (UEFI)	Newer and more advanced boot firmware. (See Firmware.)
Unsecure network	(See Open network.)
Viruses	Programs designed to spread from host to host, just like real viruses. An infected app or file has to be started by a user for a virus to activate. Viruses can turn on a webcam, record keystrokes and site visits, steal data, corrupt files, and hijack email accounts.
Virtual Private Network (VPN)	Technology that encrypts the traffic coming out of your device or site. Even if a hacker captures your data, they won't be able to read it or decrypt it. A VPN is an encrypted tunnel set up between two or more sites. All traffic in the tunnel is unreadable and useless to anyone who might try to intercept the traffic. VPN traffic is encrypted at the originating site and decrypted at the receiving site. VPNs make it much more difficult for hackers to make eavesdropping attacks, man-in-the-middle attacks, and replay attacks. This is especially useful if public wifi is your only connection option. VPNs can be hardware based or software based.
VPN hardware device	Device specifically designed to create VPNs or network devices with added VPN functionality. They include VPN concentrators, routers, and firewalls.
VPN, host-to-host	Host-to-host VPN is when two remote users need to connect securely to each other. This user-to-user or device-to-device interaction doesn't require a complex encryption solution. Both host devices use VPN software. Traffic that needs to traverse the internet is encrypted by the sender's VPN and then decrypted by the receiver's VPN. (See VPN, or virtual private network.)
VPN, host-to-site	Host-to-site VPN is when a remote user needs to connect securely to a site. The host device uses VPN software while the site uses a VPN device or software to protect its internal network. Traffic that needs to traverse the internet is encrypted by the sender's VPN and then decrypted by the receiver's VPN, which then receives the

	traffic if the host is the receiver, or routes it to its intended recipient if the site is the receiver. (See VPN.)
VPN, site-to-site	Site-to-site VPN is when two sites connect across an existing internet connection with a VPN device. Each site's internal network traffic is unencrypted. Traffic that needs to traverse the internet is encrypted by the origination site's VPN and then decrypted by the receiving site's VPN device, which then routes the traffic to its intended recipient. (See VPN.)
Whois Lookup tool	Online tool that shows who owns a site.
Windows Defender	Pre-loaded anti-malware software from Microsoft. Windows Defender protects your computer against viruses and malware for free. It also integrates with the built-in firewall that comes with the Windows operating system and is enabled by default. However, Windows Defender does not perform VPN functions.
Wiretapping	Covertly intercepting phone-based and internet-based telecommunications. Any type of traffic is a target: computer transmissions, phone calls, texts, social media updates, and fax transmissions. This can be done legally with a warrant, or it can be used illegally in cyberattacks. Wiretapping only involves listening. It cannot alter or stop data. Wiretapping is also known as lawful interception.
Worms	Viruses that start themselves after identifying system weaknesses. They don't rely on apps or files. Unlike viruses, worms can be controlled remotely.